

Name: _____

Date: _____

FRACTIONS, DECIMALS AND PERCENTAGES, (PERCENTAGE INCREASE AND DECREASE)

LO: I can increase or decrease an amount by a given percentage.

Increasing and decreasing by a %

A **percentage increase** means adding that percentage of the original. A **percentage decrease** means subtracting it.

Multiplier method: Increase by 20% multiply by 1.20.
Decrease by 15% multiply by 0.85.

Key methods

●	Increase £200 by 10% = £220	$\times 1.10$
●	Decrease £80 by 25% = £60	$\times 0.75$
●	Multiplier for +30% is 1.30	$1 + 0.30$
●	Increase by 20% means $\times 20$	should be $\times 1.20$
●	Decrease by 10% $\times 0.10$	should be $\times 0.90$

Key idea

Increase: multiply by $(1 + \text{rate})$. Decrease: multiply by $(1 - \text{rate})$.

New = Original $\times$ multiplier; e.g.

+20% $\times$ 1.20

Multiplier = $1 \pm$ percentage as decimal

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - percentage increase

Increase £250 by 20%.

$$20\% \text{ of } 250 = 50$$

$$250 + 50 = 300$$

$$\text{Or: } 250 \times 1.20 = £300$$

= £300

Add the percentage to the original.

Example 2 - percentage decrease

Decrease £180 by 15%.

$$15\% \text{ of } 180 = 27$$

$$180 - 27 = 153$$

$$\text{Or: } 180 \times 0.85 = £153$$

= £153

Subtract the percentage from the original.

Example 3 - multiplier method

A house worth £200,000 increases by 5%. Find new value.

$$\text{Multiplier} = 1 + 0.05 = 1.05$$

$$200000 \times 1.05 = 210000$$

$$\text{New value} = £210,000$$

= £210,000

The multiplier method does it in one step.

Example 4 - sale price

A jacket costs £85. It is reduced by 40% in a sale. Find the sale price.

$$\text{Multiplier} = 1 - 0.40 = 0.60$$

$$85 \times 0.60 = 51$$

$$\text{Sale price} = £51$$

= £51

Multiply by 0.60 because you pay 60% of the original.

YOUR TURN – PRACTICE (deliberate practice)

1. Percentage increase

Increase each amount by the given percentage.

- a) £100 by 10%
- b) £60 by 25%
- c) £200 by 15%
- d) £450 by 20%

2. Percentage decrease

Decrease each amount by the given percentage.

- a) £80 by 10%
- b) £120 by 25%
- c) £500 by 30%
- d) £240 by 5%

3. Multiplier method

Write the multiplier and calculate the new amount.

- a) £300 increased by 12%
- b) £150 decreased by 8%
- c) £90 increased by 35%
- d) £400 decreased by 17.5%

4. Word problems

Find the new amount.

- a) Rent £800/month increases 5%
- b) Car worth £12,000 depreciates 20%
- c) Population 50,000 grows 3%
- d) Price £45 + 20% VAT

5. Mixed increase/decrease

Calculate each.

- a) £160 increased by 30%
- b) £95 decreased by 40%
- c) 500 g increased by 12%
- d) £72 decreased by $33\frac{1}{3}\%$
- e) £1000 increased by 7.5%

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Increase by 20% means multiply by 1.20 ☐

Correct answer: _____

Reason: _____

b) Decrease by 10% means multiply by 0.10 ☐

Correct answer: _____

Reason: _____

c) A 50% decrease halves the amount ☐

Correct answer: _____

Reason: _____

d) Increase by 100% doubles the amount ☐

Correct answer: _____

Reason: _____

e) A 25% increase then 25% decrease returns to original ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A house is worth **£250,000**. It increases by **10%** one year, then decreases by **10%** the next. Is it back to £250,000? Explain why or why not.

Expression: _____

Simplified: _____

b) After a **20% discount**, a laptop costs **£560**. What was the original price? (Hint: £560 is 80% of the original.)

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

